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ENGINEERING



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Citizen-Driven Community Infrastructure Damage Assessment and Resolution Platform

CSA0905-PROGRAMMING IN JAVA

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Problem Background

Application Domain

- Smart City Management
- Infrastructure Monitoring
- Civic Issue Reporting

Current Scenario

Infrastructure issues such as potholes, broken streetlights, drainage blockages, and water leaks are often reported through manual or fragmented complaint systems, leading to delayed repairs, poor tracking, and limited transparency.

Need for the Project

A centralized platform is needed to enable citizens to report infrastructure damage, prioritize complaints based on severity, track repair progress, and improve communication between citizens and local authorities for faster issue resolution.

Target Users

- Citizens and local residents
- Municipal and local government authorities
- Public works and maintenance departments

Problem Statement

Problem Definition

Current infrastructure complaint systems are inefficient, making it difficult for citizens to report issues and for authorities to prioritize and resolve them in a timely manner.

Evidence / Statistics

Manual complaint handling often results in delayed repairs, duplicate complaints, poor tracking, and reduced transparency, leading to lower public satisfaction and inefficient maintenance.

Limitations of Current Systems

- Manual or fragmented complaint registration.
- No automatic prioritization of critical issues.
- Limited complaint status tracking and citizen notifications.
- Poor coordination between citizens and maintenance departments.

Expected Impact

- Faster reporting and resolution of infrastructure issues.
- Improved transparency through real-time complaint tracking.
- Better prioritization and resource allocation for maintenance.



Objectives

Primary Objective

To develop a Java-based platform that enables citizens to report infrastructure damage, prioritize complaints, and track their resolution efficiently.

Measurable Objectives

- Enable citizens to submit infrastructure complaints with location, description, and image details.
- Categorize and prioritize complaints based on severity and public impact.
- Store and manage complaint records securely using a MySQL database.
- Allow authorities to update repair status from Pending to Resolved.
- Provide citizens with complaint tracking and status notifications.

Project Scope

- Covers reporting of public infrastructure issues such as roads, streetlights, drainage, and water leaks.
- Provides separate interfaces for citizens and administrators.
- Stores complaint details securely in a MySQL database.
- Prioritizes complaints based on severity and public impact.
- Tracks repair progress from complaint registration to resolution.



Literature Survey

Recent Papers

1. AI-Based Road Damage Detection Using Deep Learning (2024)
2. Smart City Infrastructure Monitoring Systems (2023)
3. Crowdsourced Civic Issue Reporting Platform (2023)
4. GIS-Based Urban Infrastructure Damage Assessment (2024)

Existing Work	Strength	Limitation
AI Road Damage Detection	Accurate detection	No citizen reporting
Smart City Monitoring	Real-time monitoring	Limited public participation
Crowd-Sourced Reporting	Easy issue reporting	No severity prioritization
GIS Damage Mapping	Location-based analysis	No repair tracking

Research Gap: Existing infrastructure complaint systems often lack integrated complaint prioritization and end-to-end resolution tracking. A centralized platform is needed to improve transparency, communication, and timely maintenance of public infrastructure.

Existing vs Proposed System

Existing System	Proposed System
Manual inspections	Citizen-based reporting
Delayed issue detection	Real-time reporting
No severity prioritization	AI-based severity scoring
Limited tracking	Live repair status tracking

Advantages

- Easy reporting of infrastructure issues.
- Faster complaint processing and resolution.
- Automatic prioritization of critical complaints.
- Real-time complaint status tracking.
- Improves transparency and accountability.
- Reduces manual paperwork.
- Better communication between citizens and authorities.
- Efficient management of maintenance resources.

Requirements

Functional Requirements

- Report infrastructure issues
- Upload geotagged photos
- Prioritize severe issues
- Track repair status

Non-Functional Requirements

- User-friendly interface
- Secure and reliable
- Scalable performance
- Fast response time

Hardware Requirements

- Computer
- Smartphone with GPS & Camera
- Internet connection

Software Requirements

- Programming Language: Java
- Frontend: HTML, CSS, JavaScript
- Backend: Java (Servlets/JSP)
- IDE: Eclipse / IntelliJ IDEA / NetBeans
- Database: MySQL
- Database Connectivity: JDBC
- Web Server: Apache Tomcat
- Database Tool: MySQL Workbench

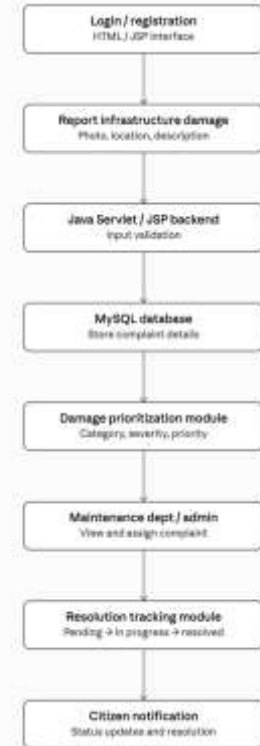
Dataset / APIs

- GPS Location API : OpenStreetMap, Google Maps API
- Municipal infrastructure datasets and Image dataset for damage classification

Architecture

Workflow

- User Login: Citizens register or login to access the complaint portal.
- Complaint Submission: Users submit infrastructure damage details along with location, description, and image.
- Data Validation: The system verifies the entered information and stores it in the MySQL database.
- Damage Prioritization: Complaints are categorized and assigned a priority level based on severity and public impact.
- Department Assignment: Prioritized complaints are forwarded to the appropriate maintenance department.
- Repair Progress Tracking: Administrators update the complaint status from Pending → In Progress → Resolved.
- Citizen Notification: Users receive status updates and can track the progress until the complaint is resolved.
- Complaint Closure: The complaint is marked as completed and stored for future reference.





Module Descriptions

Module 1: Citizen Reporting Module

- Provides user registration and login functionality.
- Allows citizens to report infrastructure damage.
- Accepts damage category, location, description, and image.
- Validates and stores complaint information in the database.
- Generates a unique complaint ID for tracking.

Module 2: Damage Prioritization Module

- Retrieves newly submitted complaints from the database.
- Categorizes complaints based on infrastructure type.
- Evaluates damage severity and public impact.
- Assigns priority levels (Low, Medium, High, Critical).
- Forwards prioritized complaints to the maintenance department.

Module 3: Resolution Tracking Module

- Displays assigned complaints to administrators.
- Updates repair progress through different status levels.
- Maintains repair history and completion records.
- Notifies citizens whenever the complaint status changes.
- Closes the complaint after successful resolution and archives the record.

Planning

Work Breakdown Structure (WBS)

1. Project Planning

- Problem Identification
- Requirement Analysis
- Literature Survey
- System Design

2. User Interface Development

- Login and Registration
- Citizen Dashboard
- Complaint Submission Form
- Admin Dashboard

3. Backend Development

- User Authentication
- Complaint Management
- Damage Prioritization
- Resolution Tracking

4. Database Development

- Database Design
- MySQL Database Creation
- JDBC Connectivity
- Data Management

5. Testing

- Unit Testing
- Integration Testing
- Bug Fixing
- System Testing

6. Deployment and Documentation

- Application Deployment
- User Documentation
- Project Report Preparation
- Final Presentation

Planning

Risk Analysis

Risk	Impact	Mitigation
Database connection failure	High	Perform regular database backups and connection testing.
Incorrect complaint data	Medium	Validate all user inputs before storing data.
Delays in development	Medium	Follow the project schedule and review progress weekly.
Software bugs	High	Conduct unit and integration testing throughout development.
Unauthorized system access	High	Implement secure login and user authentication.
Data loss	High	Maintain regular database backups and recovery procedures.

Expected Outcomes

Deliverables

- Java-based web application for infrastructure complaint management.
- Citizen and administrator login modules.
- Complaint reporting, prioritization, and resolution tracking system.
- MySQL database with JDBC connectivity.
- Project documentation, source code, and user manual.

Benefits

- Enables citizens to report infrastructure issues easily.
- Improves complaint prioritization and faster issue resolution.
- Enhances transparency through complaint status tracking.
- Reduces manual paperwork and improves maintenance efficiency.
- Strengthens communication between citizens and local authorities.



References (IEEE)

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